Ethics, Communication & Organization – Chapter Summary

Principles • Consent • End of Life • Research • Quality

ICU / EDIC Board Review • MCQ-Oriented

1. Four Principles of Bioethics

- **Autonomy:** respect informed, capacitous choices
- **Beneficence:** act in the patient's best interest
- **Non-maleficence:** "first, do no harm"
- **Justice:** fair allocation of resources
- Others: dignity, truthfulness, confidentiality, proportionality
- **Double effect:** giving opioids to relieve distress that may hasten death is acceptable if intent is comfort, dose proportionate

2. Capacity & Consent

- **Capacity** is decision-specific & time-specific
- **Assessment:** understand, retain, weigh, communicate the decision
- **Valid consent:** capacity + information + voluntary
- **Emergency:** treat in best interests if no capacity & no surrogate
- **Surrogate decision-maker:** substituted judgment (what patient would want) > best interest standard
- **Advance directive** valid if made with capacity, applicable to situation
- Competent adult may **refuse** life-sustaining treatment (incl. blood – Jehovah's Witness)

3. Withholding & Withdrawing

- **Ethically equivalent** (no moral difference between withholding and withdrawing)
- **Medically futile / non-beneficial** treatment: no obligation to provide
- **Time-limited trial:** agreed goals + timeframe, then reassess
- Withdrawal ≠ abandonment: continue **comfort care**
- Document decisions, rationale & discussions clearly
- **Most ICU deaths** follow decisions to limit life-sustaining therapy

4. End-of-Life Care Practice

- **Symptom control:** opioids for dyspnea / pain, benzodiazepines for anxiety, glycopyrrolate for secretions
- **Terminal extubation** vs terminal weaning: both acceptable; premedicate
- **Stop NMB** before withdrawal (masks distress)
- Discontinue non-comfort interventions (labs, monitoring alarms)
- Deactivate ICD
- **Euthanasia / physician-assisted death:** legally distinct; jurisdiction-specific

5. Family Communication

- **VALUE:** Value, Acknowledge emotions, Listen, Understand the patient as a person, Elicit questions → ↓ PTSD, anxiety (Lautrette)
- **SPIKES** for breaking bad news: Setting, Perception, Invitation, Knowledge, Emotions, Strategy/Summary
- **Shared decision-making** model
- Proactive family conferences within 72 h
- **Conflict:** second opinion, ethics consult, mediation, legal last resort
- Interpreters for language barriers (not family members)

6. Post-ICU Outcomes (PICS)

- **PICS:** physical (ICU-acquired weakness), cognitive impairment, psychological (anxiety, depression, PTSD)
- **PICS-F:** family members affected
- **Prevention:** ABCDEF bundle, ICU diaries, follow-up clinics
- **Burnout** in ICU staff: high; ↓ with team support, debriefing

7. Resource Allocation & Triage

- **Triage:** maximize benefit; criteria transparent & consistent
- Not based on social worth, race, disability
- **Pandemic / mass casualty:** SOFA-based triage protocols (controversial)
- **Rationing:** explicit vs implicit
- **Utilitarian** (greatest good) vs **egalitarian** (equal access)
- ICU admission: benefit, patient wishes, prognosis

8. Confidentiality & Disclosure

- Breach permitted: risk of serious harm to others, legal requirement, public health
- **Open disclosure / duty of candour** after adverse events: apologize, explain, prevent recurrence
- **Medical errors:** report via incident systems (non-punitive "just culture")
- Truth-telling even if family requests otherwise (respect patient's wishes)

9. Research Ethics

- **Declaration of Helsinki**, Good Clinical Practice (GCP)
- **Informed consent** – in ICU often **deferred consent** / waiver (emergency research) approved by ethics committee
- **Equipoise:** genuine uncertainty needed
- **Vulnerable population:** extra protection
- **Data Safety Monitoring Board** for interim analyses

10. Evidence-Based Medicine

Term	Meaning
Sensitivity	TP / (TP + FN) – rules OUT (SnNout)
Specificity	TN / (TN + FP) – rules IN (SpPin)
PPV / NPV	Depend on prevalence
LR+	Sens / (1 – Spec)
NNT	1 / ARR
Type I error (α)	False positive (usually 0.05)
Type II error (β)	False negative; power = 1 – β (80–90%)

- **Fragility index:** number of events to lose significance
- **Intention-to-treat** preserves randomization
- **Bayesian analysis:** probability of benefit

11. Quality & Safety

- **Donabedian:** structure, process, outcome
- **Bundles & checklists** (CLABSI: hand hygiene, barrier precautions, chlorhexidine, avoid femoral, remove early)
- **SMR** = observed / expected deaths (APACHE, SAPS)
- **Handover:** structured (SBAR / I-PASS)
- **Root cause analysis**, PDSA cycles
- **Swiss cheese model** (Reason)
- **Rapid response teams / outreach** for deteriorating ward patients

12. Severity Scoring

Score	Use
APACHE II	First 24 h, 12 physiologic variables + age + chronic health
SAPS II / III	Admission mortality prediction
SOFA	Organ dysfunction (serial)
MPM	Mortality probability
NEWS2	Ward deterioration

- Scores for **populations**, not individual decisions

13. ICU Organization

- **Intensivist-led (closed) units** ↓ mortality
- Nurse : patient ratio 1:1 (level 3) / 1:2 (level 2)
- **Tele-ICU** support
- **Infection control:** hand hygiene, isolation, antimicrobial stewardship
- Teamwork & **crisis resource management** (CRM)

14. High-Yield MCQ Pearls

- ☑ Withholding = withdrawing ethically
- ☑ Capacity is decision & time specific
- ☑ Substituted judgment > best interests
- ☑ Competent refusal must be respected
- ☑ Double effect justifies opioids at end of life
- ☑ Stop NMB before withdrawal
- ☑ VALUE ↓ family PTSD
- ☑ Deferred consent in emergency research
- ☑ Sensitivity rules out, specificity rules in
- ☑ NNT = 1 / ARR
- ☑ Severity scores for groups, not individuals
- ☑ Closed ICU model ↓ mortality

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